

DS	Session	Tree 1	Question Information	Level 1	Challenge 66
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Question description

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director. Your task is to process q queries of the form: who is employee x 's boss k levels higher up in the hierarchy?

Input

The first input line has two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director.
The next line has $n-1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss.
Finally, there are q lines describing the queries. Each line has two integers x and k : who is employee x 's boss k levels higher up?

Output

Print the answer for each query. If such a boss does not exist, print -1 .

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq e_i \leq i-1$
- $1 \leq x \leq n$
- $1 \leq k \leq n$

Logical Test Cases

Test Case 1

INPUT (STDIN)

```
5 3
1 3 2 2
4 2
1 2
1 3
```

EXPECTED OUTPUT

```
1
-1
-1
```

Test Case 2

INPUT (STDIN)

```
5 3
1 1 3 3
4 1
4 2
4 3
```

EXPECTED OUTPUT

```
3
1
-1
```

Mandatory Test Cases

Test Case 1

KEYWORD

```
void link(int i,int j)
```

▼ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
2

Test Case 2
TOKEN COUNT
420

Test Case 3
NLOC
50